

A Non-Ultrametric Example with Positive Snowflake Self-Embedding Parameter

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Source Metadata

This packet concerns arXiv:1206.3774, *Embeddability of snowflaked metrics with applications to the nonlinear geometry of the spaces L_p and ℓ_p for $0 < p < \infty$* , by Fernando Albiac and Florent Baudier. The arXiv page records the paper as submitted on 2012-06-17, revised on 2012-10-02, and later published in *Journal of Geometric Analysis* 25 (2015), no. 1, 1–24.

The relevant open question appears near the end of the paper as Question q3, after the definition

$$\beta_{\mathcal{M}} = \sup\{t \geq 0 : (\mathcal{M}, d) \text{ Lip}(\mathcal{M}, d^{1-t})\}.$$

Here $(\mathcal{M}_1, d_1) \text{ Lip}(\mathcal{M}_2, d_2)$ means that there is a one-to-one map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and constants $A, B > 0$ such that

$$A^{-1}d_1(x, y) \leq d_2(f(x), f(y)) \leq Bd_1(x, y) \quad (x, y \in \mathcal{M}_1).$$

Question q3 asks for a description of the complement of $\mathbf{NS}_{\top}$ and, in particular, whether there are non-ultrametric metric spaces with $\beta_{\mathcal{M}} > 0$.

Result

The result below answers the existential part of Question q3 on the literal reading of the question. It also records a simple family contained in $\mathbf{CNS}_{\top}$.

Proof Intuition

The obstruction in self-snowflake embeddings is that the snowflake metric d^s , $0 < s < 1$, changes small and large scales at different rates. A bounded uniformly discrete metric space has no arbitrarily small or arbitrarily large nonzero distances. On the compact interval of possible distances $[a, b]$, the two functions r and r^s are comparable by fixed constants. Therefore the identity map is already a Lipschitz embedding from (M, d) into (M, d^s) . Choosing a bounded uniformly discrete metric that violates the strong triangle inequality gives the requested non-ultrametric example.

Lemma 1 (bounded uniformly discrete spaces self-embed into snowflakes). *Let (M, d) be a metric space for which there are constants $0 < a \leq b < \infty$ such that*

$$a \leq d(x, y) \leq b \quad \text{whenever } x \neq y.$$

Then for every $0 < s < 1$, the identity map is a Lipschitz embedding

$$(M, d) \text{Lip}(M, d^s).$$

Consequently, $\beta_M > 0$. More precisely, for every $0 < t < 1$, $(M, d) \text{Lip}(M, d^{1-t})$, so the supremum is at least 1 when β_M is interpreted on the usual snowflake range $0 < 1 - t < 1$.

Proof. Fix $0 < s < 1$ and put $r = d(x, y)$. The case $x = y$ is immediate. For $x \neq y$, $r \in [a, b]$. Since $s - 1 < 0$,

$$b^{s-1} \leq r^{s-1} \leq a^{s-1}.$$

Multiplying by r gives

$$b^{s-1}d(x, y) \leq d(x, y)^s \leq a^{s-1}d(x, y).$$

Equivalently,

$$(b^{1-s})^{-1}d(x, y) \leq d(x, y)^s \leq a^{s-1}d(x, y).$$

Thus the identity map satisfies the source paper's Lipschitz embedding inequalities with $A = b^{1-s}$ and $B = a^{s-1}$. Taking $s = 1 - t$ proves the claim about β_M . $\square$

Proposition 1 (explicit non-ultrametric example). *There is an infinite metric space (M, d) that is not an ultrametric and has $\beta_M > 0$.*

Proof. Let

$$M = \{a, b, c\} \cup \{x_n : n \in \mathbb{N}\}.$$

Define $d(x, x) = 0$, make d symmetric, and set

$$\begin{aligned} d(a, b) &= 1, & d(b, c) &= 1, & d(a, c) &= \frac{3}{2}, \\ d(x_n, u) &= 2 \quad (u \in \{a, b, c\}, n \in \mathbb{N}), & d(x_m, x_n) &= 1 \quad (m \neq n). \end{aligned}$$

We first check that this is a metric. The only triangle entirely inside $\{a, b, c\}$ that is not immediate is

$$d(a, c) = \frac{3}{2} \leq d(a, b) + d(b, c) = 2.$$

If a triangle has one point x_n and two points in $\{a, b, c\}$, the largest possible side is 2, while the sum of the other two sides is at least $2 + 1$; the side joining two core points is at most $3/2 \leq 2 + 2$. If a triangle has two points x_m, x_n and one core point, the only nontrivial inequality is $2 \leq 1 + 2$. Triangles involving only the x_n 's are equilateral. Hence d is a metric.

All distinct distances lie in $[1, 2]$, so Lemma 1 applies and $\beta_M > 0$. However, M is not ultrametric, because

$$d(a, c) = \frac{3}{2} > \max\{d(a, b), d(b, c)\} = 1.$$

This gives the requested non-ultrametric example. $\square$

Verification Notes and Limitations

The proof is purely metric and uses no external embedding theorem. It depends only on the comparability of r and r^s on the interval of nonzero distances of a bounded uniformly discrete metric space.

This packet should be read as a full solution to the yes/no clause of Question q3. It is not a full classification of $\mathbf{CNS}_\top$. If the intended problem was meant only for unbounded metric spaces, Banach-space metrics, spaces without isolated points, or examples not explained by bounded uniform discreteness, this packet does not settle that stronger interpretation.

Bounded Novelty Check

Before promotion, the following bounded checks were performed on 2026-07-18. The local run indexes were searched:

- `registry_index.tsv`;
- `solutions/index.tsv`;
- `attempts/index.tsv`;
- `proof_gaps/index.tsv`.

The search terms were:

- `1206.3774`, `beta_M`, and β_M ;
- `other than ultrametrics`;
- `bounded uniformly discrete` and close variants.

The local source and deterministic scan were also searched. No duplicate packet or prior run answer to this exact observation was found.

External web searches were also run for phrases around β_M , snowflaked self-embeddings, ultrametrics, and bounded uniformly discrete spaces. These found the source arXiv record and nearby snowflake-characterization literature, but no exact recorded answer to this literal existential clause. Novelty confidence is therefore moderate rather than absolute, mainly because the observation is elementary and may have been considered implicit.

Human Review Recommendation

Human review should focus on whether Question q3 was intended to exclude bounded uniformly discrete spaces implicitly. On the literal definition printed in the source, the proof above gives a valid non-ultrametric example with $\beta_M > 0$.

References

- [1] Fernando Albiac and Florent Baudier, *Embeddability of snowflaked metrics with applications to the nonlinear geometry of the spaces L_p and ℓ_p for $0 < p < \infty$* , arXiv:1206.3774, Journal of Geometric Analysis 25 (2015), no. 1, 1–24.